

Nicolás Saldarriaga Carvajal

FULL-STACK DEVELOPER · FINANCIAL DATA · PYTHON
/ FLASK / POSTGRESQL / AZURE

I design, build and run the software that finance teams decide with. In 2026 I built and now operate two production web platforms for Prisier, working with a talented team of pricing consultants at a firm that prioritizes AI-first strategies. Before that, I built data tools for the IDB Treasury and for IFAD's Latin America portfolio.

I know how treasury, portfolio and risk teams use their data, and I build the systems that serve it.

11 days

To design, test and deploy Prisier One on Microsoft Azure (BOps took ~4 months to design, test, build and deploy)

409

Commits · Jul–Oct 2026

~69k

Lines of Python in production

~740

Automated tests

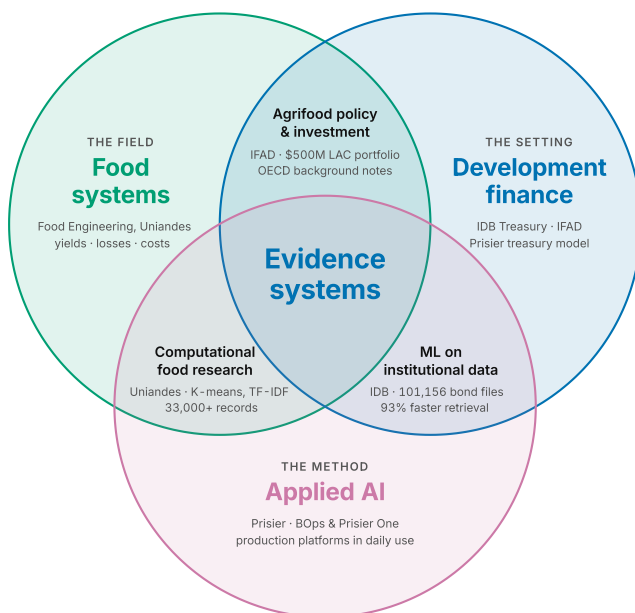
101,156

IDB investment files classified with ML

\$500M+

IFAD portfolio I automated tracking for

1) Profile · Where finance, data and software overlap



Evidence architect for food & development finance

Evidence that moves policy and capital.

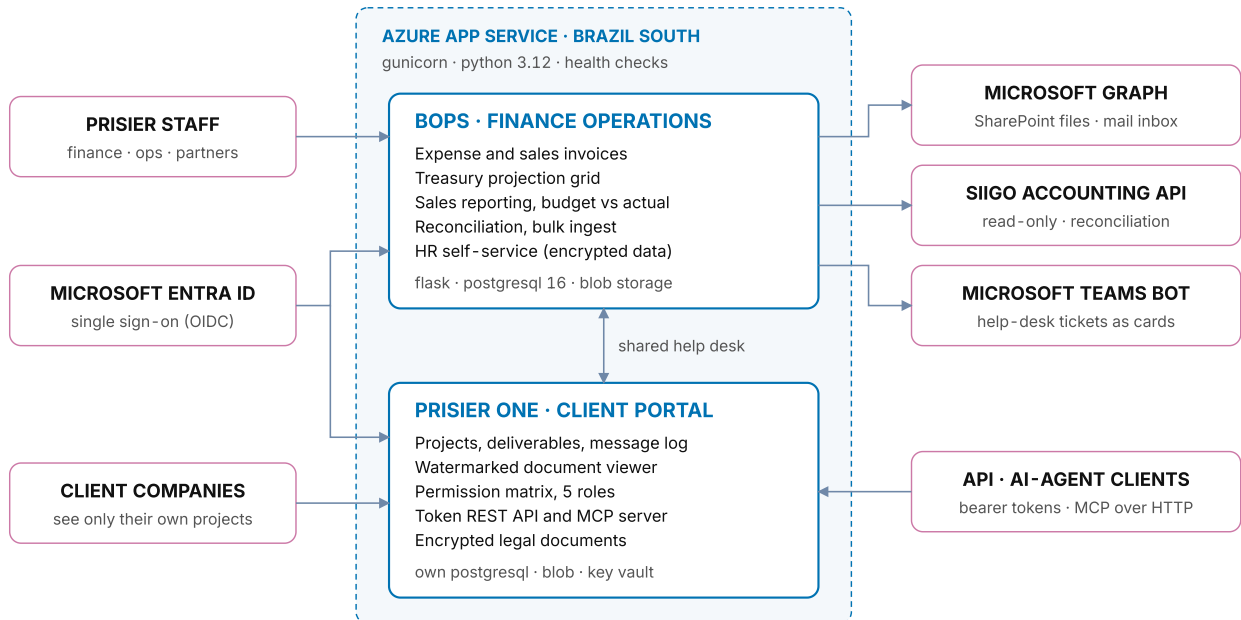
I turn the messy information institutions sit on into evidence they can decide with. I learned the questions inside IFAD, the IDB and the OECD, and I build the systems, now **with AI**, that answer them.

- **Food systems** is the field
- **Development finance** is the setting
- **Applied AI** is the method
- **Evidence systems** is what I deliver

2) Prasier · Two production platforms on Azure Feb 2026 – present · Bogotá

Prasier is a pricing and revenue-management consultancy. I designed, built, deployed and maintain both of its internal platforms. BOps runs the firm's finances; Prasier One is where clients receive their deliverables.

Architecture 2 apps · isolated data · 1 sign-in · 1 help desk



BOps · back-office finance

241 commits · ~42k lines · 56 models

- **Invoices:** expense and sales invoices from upload, bulk zip import or a monitored mailbox, with duplicate checks, extraction and an approval step.
- **Treasury:** a weekly, monthly and yearly cash projection. Every cell is computed from invoice and debt data, and the balance is anchored to real bank balances.
- **Financial model:** generates an Excel workbook with live formulas and three cash scenarios (committed, weighted, full pipeline) from a read-only database transaction.
- **Reconciliation:** a one-way sync with the accounting system. It changes a record only when the match is unambiguous, and every change is audited.
- **Dashboards:** sales by client and month, budget vs actual, pipeline. Manual overrides are flagged when they differ from the computed value.

Prasier One · client portal

168 commits · ~28k lines · ~740 tests

- **Access control:** one permission matrix is the single source of truth, and a test restates it independently. The app refuses to start if a client could ever reach internal tools.
- **Secure viewer:** documents are served in a sandboxed frame with a watermark naming the viewer and the time. Permission is rechecked on every load.
- **API and MCP:** a token REST API and an MCP server let staff publish deliverables from scripts and AI agents.
- **Operations:** scheduled maintenance windows, release notices, bulk account creation and audited deletes behind three confirmations.

Source code is private and belongs to Prasier SAS. A live walkthrough of both platforms is available on request.

3) Results · What changed, measured Measured on real company data

Metric	Before → after	What I built and how it was measured
Cash cover of sales	~11 → ~30 days	Jan–Sep 2026 vs. the same period of 2025: sales roughly doubled, and average month-end cash in the same bank accounts grew about 6×, so cash per peso billed is almost 3× higher. Since BOps, every sales and expense invoice sits in one place with its collection or payment status, and treasury is projected week by week on real bank balances. Approximate figures, adjusted for sales growth.
Wrong invoice totals	19% → 5.3%	Replaced a vision-AI reader with a deterministic PDF parser that accepts a value only when two independent methods agree. Held-out test: 281 of 778 confirmed invoices. Sales invoices now read 90% exact.
Help-desk wrong answers	18% → 7%	Rebuilt the help-desk bot as a search engine over the knowledge base that escalates to a person when unsure. Correct answers went from 54 to 86, with 0 of 25 false positives.
Invoice batch time	–30 min	Saved per 250-invoice batch (about 7 s per invoice) by removing the AI model's rate-limit pause.
Cash-timing accuracy	227 / 395	Expense rows moved to their correct payment week in the treasury projection, with the total unchanged.
Document load time	13.2 s → 0.67 s	A 100-page deck in the client portal.

4) Security hardening · built for financial and personal data OWASP ASVS Level 2 review

Identity and access

- Microsoft Entra single sign-on for staff. Users are matched on an immutable ID and accounts are never created automatically.
- Roles live in the app's own database: one permission matrix, checked on every request and restated in tests.
- Separate rights for exporting data, scoped per partner.

Data protection

- Personal and payroll data encrypted at the column level with a dedicated key. Without the key the module refuses to serve, so it never falls back to plain text.
- Legal documents stored with envelope encryption (AES-GCM).
- Client data kept on separate database, storage and key-vault resources. Both apps hosted in Brazil South for Latin American data residency.

Application and integrations

- Strict content security policy with zero inline scripts; I moved 128 inline handlers to remove the last exception.
- API tokens expire, can be scoped to one project and are stored only as hashes. The API is off by default and every action is audited.
- Least privilege on integrations: mailbox access is limited to one mailbox (verified: allowed there, refused elsewhere), the accounting API is read-only, and the help desk queries the database in read-only transactions.

Control

In place

Pre-deploy audit	Every route checked for an access guard before each release; must report zero unguarded routes.
Abuse protection	Rate limiting on sign-in and password reset, account lockout, CSRF protection, idle timeouts, forced sign-out when a password changes.
Uploads	File type checked from content rather than the extension, zip limits against decompression bombs, image-only attachments with SVG refused.
Audit trail	An append-only security log, an admin audit viewer and a compliance view of every account's permissions.
Resilience	Health checks, auto-heal, metric alerts, 35-day geo-redundant database backups, storage versioning, feature flags and kill switches per module.

5) Analytics · Pricing for a national apparel retailer Prisier client · Mar 2026

For a Prisier client with more than 230 stores in four countries, I analysed a full year of sales: **6.2M** transaction lines in a **1.3 GB** file, in Python (pandas, NumPy, matplotlib).

- Normalised mixed currencies and stores that record order totals instead of per-unit prices.
- Computed margin per transaction and broke it down by country, store, channel, product line and month.

The client's name and figures are confidential and left out on purpose.

- Found that **2.1%** of transactions were sold below cost, and traced the losses to a few stores where 86–98% of sales carried a deep discount.
- Delivered 10 charts, written reports and a presentation.

6) Inside the IDB, IFAD and the OECD

Inter-American Development Bank · Treasury

Washington D.C. · Jun–Aug 2025

- Classified **101,156** fixed-income investment files with machine learning, cutting file retrieval time by **93%**.
- Built a central data platform with **14,987** bond definitions and **13,846** strategy records (Python, VBA/Excel). Manual processing fell **60%**, and portfolio reporting became a repeatable process.

Python

ML classification

VBA

Fixed income

IFAD · Latin America and Caribbean

Rome · Jul 2024–Jan 2025

- Automated portfolio information flows and impact tracking for **30+** projects in a **US\$500M+** portfolio. Request processing time fell **85%**, and the work informed a **55%** increase in thematic work across the portfolio, most of it on business development for rural populations.
- The same tools became a covenant-tracking system with **35+** KPIs per project, adopted across the region.
- Set quality-control protocols for 620+ knowledge products, including a new KPI reporting system. The department adopted the system at the time.

Automation

KPI monitoring

Portfolio data

OECD · Trade and Agriculture Directorate

Bogotá · Feb–Dec 2026

- Co-organised two national workshops between Colombia's Ministry of Agriculture and the OECD in Bogotá (July 2026), with about **90** participants from ministries, statistical agencies, the private sector, FAO, the IDB and ECLAC. Also supported four expert focus groups across six countries.
- Wrote two background notes on Colombia, about **15,000** words each, both accepted: one on measuring food loss and waste, one on equal opportunities in food systems.
- Mapped Colombia's family farm unit for every municipality in Python, as a geospatial input to the policy analysis.

Stakeholder engagement

Policy analysis

Geospatial

Universidad de los Andes

Bogotá · Aug 2023–Jun 2024

- Mixed-methods impact study of **1,774** community initiatives and **33,000+** development records in post-conflict territories.
- Used K-means clustering and TF-IDF to find investment gaps.

K-means

TF-IDF

Mixed methods

7) Role fit · Requirements and evidence SRM / RMG · Developer (PEC)

The role asks for	Where I have done it
Full-stack web apps	BOPs and Priser One: designed, built, deployed and maintained as the sole developer, both in daily production use.
Interactive dashboards	Treasury projection grid, sales and budget dashboards, and a shared chart engine checked for accessibility. Power BI before that.
APIs	Token REST API and MCP server (Priser One); clients for the Siigo accounting API, Microsoft Graph and Teams (BOPs).
Databases	PostgreSQL schemas (56 models in BOPs), idempotent migrations, read-only reporting transactions; the IDB bond and strategy data platform.
Version control	Git and GitHub for all work: 409 commits across both platforms in under three months.
Code reviews	I review every change before it ships, with test suites and a pre-deploy route audit. I'm ready to join a team review process.
Risk and treasury domain	IDB Treasury fixed-income data, IFAD covenant monitoring, and Priser's treasury, collections and reconciliation modules.

8) Stack

Backend	Python 3.12, Flask, SQLAlchemy, Gunicorn, REST, MCP
Frontend	Jinja templates, JavaScript, HTML/CSS, a custom chart engine and design system
Data	PostgreSQL, SQL, pandas, NumPy, openpyxl, Excel/VBA, Power BI
Cloud	Azure App Service, Azure Database for PostgreSQL, Blob Storage, Key Vault, Entra ID, Bot Service, Microsoft Graph
Methods	ML classification, K-means, TF-IDF, BM25 retrieval, geospatial analysis, LLM applications, AI-assisted engineering
Languages	Spanish (native), English (fluent), French (conversational)

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Bogotá, Colombia